Sunny Chang Lu

Shanghai, China | luchang1106@icloud.com | +86 138-1776-0040 | snylur.github.io | github.com/Snylur

Education

Shanghai Starriver Bilingual School Shanghai, China

Class of 2027 | *GPA: 4.770 (weighted)* | *TOEFL: 116*

AP Research (2026–27); 11 of 12 AP exams scored 5.

Audited CIS 160 and CIS 121 at University of Pennsylvania; COS 126 at Princeton University.

Summer Programs: Ross '26 — six-week, selective proof-based mathematics program

ESPR '25 — European Summer Programme on Rationality (game theory, cryptography, mathematical logic)

Yau Math Camp '25 | PACT Year-Round '21, '22 | Yingcai Talent Program '25 (SJTU CS; advisor Yu Yong)

Research

Kempe's Universality Theorem on Restricted Topologies 2025–Present

- Proved that linkages remain universal when restricted to crossing-free, bipartite graphs of maximum degree three.
- Built interactive Desmos constructions of each gadget: Desmos collection 1, Desmos collection 2.

HeXO is PSPACE-Hard 2025

- Proved $\mathrm{Connect}(n, 6, 2, 1)$ on a hexagonal grid PSPACE-complete by reduction from Quantified 3SAT.

Folding Polyominoes into Uniform-Thickness Rectangles 2025–2026

Tiling Squares with Polyominoes 2026

PowerPoint Vexillology *(Arglin's Cenote)* 2025–Present

Contributed to a 2,000+ member research effort constructing every flag with PowerPoint tools; supplied the SmartArt cycle construction necessary for flags of Croatia and Kyrgyzstan; presented the work in a talk on the Math Club channel

Awards & Qualifications

Mathematics

Silver Medal, Distinction and Best in School, British Mathematical Olympiad 2026

STEP II: 115/120 (S); STEP III: 102/120 (S), Sixth Term Examination Papers (STEP) 2026

15th Overall Individual and 11th Theme Round, Harvard–MIT Mathematics Tournament 2023

also 18th Combinatorics (2025) and 23rd Geometry (2026)

AIME: 12/15 (2025, 2026); AMC 12: 138/150 (2026); AMC 10: perfect score, top 1% (2023) 2023–2026

1st Individual and 1st Team, Stanford Math Tournament 2025

1st Individual and 1st Team (2024); 2nd Individual, High Distinction (2025), Yale MMATHS 2024–2025

Team 3rd Overall; Individual 8th, Combinatorics 3rd, Princeton University Mathematics Competition 2026

Global Top-10 Team and National 3rd, High Distinction, Math Tournament at Berkeley 2024

National Team Gold, National Individual Top 100; Power Round I Silver, ARML Local 2023

1st Team Overall and 2nd Individual (International), Duke Math Meet; 1st Team Overall, 1st Team Challenge and Relay, 3rd Individual (China) 2025

Honorable Mention, International Mathematical Modeling Challenge 2026

Informatics & Artificial Intelligence

Silver Medal, Scientific Round, International Olympiad in Artificial Intelligence 2024

Gold Division, USA Computing Olympiad 2023

First Prize, National Olympiad in Informatics in Provinces (NOIP) 2025

CSP-J and CSP-S: First Prize, both rounds (CCF informatics olympiad) 2023

Winner, 2nd International Obfuscated Python Code Contest 2024

All-Star, Global Bronze, ACSL Programming Contest 2023

Physics & Chemistry

Global Gold; Global Top Gold (top 2%); Round 2 Silver, British Physics Olympiad 2024–2026

Global Gold (49/50), Senior Physics Challenge 2025

National Gold (2025); National Silver (2024), Physics Bowl Division 1 2024–2025

Global Gold, Cambridge Chemistry Challenge (C3L6) 2024

Linguistics, Humanities & Design

High Distinction (top 10%), International Linguistics Olympiad Regional Round 2026

John Locke Institute Essay Competition 2023–2024
Commendations in Politics and Philosophy (2023); Merit in Politics (2024)

Finalist, Best Pen & Paper Puzzle, Thinky Awards 2025

Activities & Leadership

OEIS Contributor 2025–Present
Proposed sequence A398486; contributed corrections and comments to existing entries

Presenter of 2swap's Math Club 2025–Present
Seminars on tilings, foldings, and PowerPoint constructibility; related videos have 800,000+ total views on YouTube

Open-Source Developer 2023–Present
Origami Simulator, HeXO Renderer, and miscellaneous projects

Puzzle Designer 2023–Present
Co-designed Origami Puzzles, Thinky Awards finalist | co-designed ICNP | contributor to PUZZDLE
playtester for Herd, Workworkwork, and Mondrian Restoration

Speaker, TEDxXuhui 2023

Participant, LIVE Communication Training 2025–Present

Work Experience

Worksheet Author, starmathcat *(YouTube mathematics channel)* 2026–Present

Mathematics Tutor, Cambridge STEP Exam Preparation *(five sessions)* May 2026

Freelance Translator, *Checkmating Combinations in Chess* into Chinese Aug–Dec 2024

Skills

Programming: proficient in Python, Desmos, and LaTeX | open source work with Kotlin | basic C++ and Java knowledge
Languages: fluent in English, Chinese (Mandarin), and toki pona